Khushbu Parmar

Data Scientist — Bengaluru, India

Contact Information

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Research Interests

Deep Learning, Edge Deployment, GenAi, Computer Vision, NLP

Education

Master of Technology in **Computer Science** with specialization in **Data Science** (Aug 2022 - Jun 2024)
Indian Institute of Information Technology Vadodara, India. **CPI: 7.76/10.0**

Bachelor of Engineering in **Electronicss and Communication Engineering** (Aug 2017 - Jul 2021)
Gujarat Technological University, India. **CPI : 8.59/10.0**

Work Experience

Data Scientist at **Siemens** (From Sep 2024)

Bengaluru, India

1. **Developed and deployed** high-accuracy, resource-efficient **load forecasting models (1D CNN)** using historical **load and weather data**, incorporating advanced **preprocessing**, **feature engineering**, and **normalization techniques**; applied **quantization** with **TensorFlow Lite** and generated optimized **C code** using STM32Cube's **X CubeAI** package for **STM32 microcontrollers** to enable efficient **inference** on **embedded systems**.
2. Implemented a lightweight **1D CNN** in **MicroPython** for **real-time, on-device retraining and forecasting** on constrained **hardware**, **developing all components from scratch** including **forward pass**, **backpropagation**, **normalization**, and **optimization strategies**, while ensuring **scalable** and **energy-efficient solutions** through deep integration of deep learning with Hardware environment.
3. Handled the **client independently**, gathering **requirements**, presenting **solutions**, and ensuring **successful deployment**, while collaborating with **hardware engineers** for seamless **model integration** on **deployment platforms**.

Data Science Intern at **Siemens** (July 2023 - Aug 2024)

Bengaluru, India

1. Engineered an **Operational Engineering Assistant** using **Retrieval-Augmented Generation (RAG)** with **Hybrid search retrieval** and **multi-size chunking** for faster, context-aware retrieval. Designed a secure **backend-driven vector database** with **per-user and per-file isolation** via session/thread IDs, offering multiple models **(Llama 2, Llama 3, Mistral)** for user selection, **Nemoguardrails with LangChain** for safe, controlled chatbot responses.
2. Built and integrated **APIs using FastAPI with Cassandra DB to support CRUD operations**, chat functionality, **thread management**, and **real-time data streaming**, enabling seamless interaction with external plant systems and automated error resolution workflows.
3. Created an **Automated computer vision system for Engineering process drawings identification**, using **YOLOX**, **HoughLines for line/connection detection**, and **OCR-based text recognition** to identify instruments tags , enabling accurate **digital reconstruction** of complex diagrams within target systems.

Data Science Intern at **Visione Infotech** (Feb 2022 - July 2022)

Surat, India

1. Developed and maintained dynamic **front-end features** using **Angular** and **Postman** for **Diamond Trading Cloud Software**, enhancing user workflows in **inventory**, **CRM**, **invoicing**.

Academic Projects

Birds Audio Identification with Speech Representation Learning Transformer Model : A bird audio classification system using **Xeno-Canto dataset** (105 species, 100 GB data) by fine-tuning the **DistilHuBERT Transformer**, implementing a robust preprocessing pipeline with **Librosa**, to improve generalization and outperform traditional convolutional models.

DocuFusion AI Assistant: Based on Fusion RAG : Built a multi-format **Document Retrieval System** using **Streamlit** and **Ollama**, integrating **BM25** and **VectorStore retrievers** with **QueryFusionRetriever** for improved accuracy, enabling **PDF/JSON/Text indexing** and seamless **LLM switching** (**LLaMA2, Mistral, Phi, LLaVA**).

Music genre classification with CNN: Created an end-to-end **Music genre classification system** using **Librosa** for audio feature extraction (mel-spectrograms with chunked audio processing) and a deep **CNN** architecture in **TensorFlow/Keras**, trained on the GTZAN dataset to achieve high accuracy in genre prediction.

Voice gender classification model: Built a **voice gender classification model** using **Random Forest** on a dataset of 20 acoustic features, applying **feature selection**, **correlation analysis**, and **data visualization** (Seaborn, Matplotlib) to optimize performance.

Skills

Parse skills from this section

Deep Learning: Neural Networks, CNNs, RNNs, Transformers, Object Detection & Classification, Image Segmentation, Sequence Modeling, Seq2Seq Prediction, Keras, PyTorch, TensorFlow, OpenCV, MicroPython, TFLite, CUDA, YOLO, NLTK

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Ensemble Methods (Random Forest, XGBoost), Clustering, Dimensionality Reduction, Feature Engineering, Hyperparameter Tuning, Transfer Learning, Scikit-Learn, Python, MATLAB, Pandas, NumPy, Matplotlib, Seaborn

Generative AI (GenAI): Large Language Models, Retrieval-Augmented Generation (RAG), Prompt Engineering, Few-shot & Zero-shot Learning, Fine-tuning, Embedding Generation & Search, Multimodal Generation, AI Safety, Conversational Agents, LangChain, LlamaIndex, HuggingFace Transformers, FAISS, Weaviate, Streamlit, LLaMA, DistilBERT, DistilHuBERT